Due to the optimality and losslessness of Huffman coding (Huffman, 1952), the compression protocol ensures optimal compression while preserving losslessness. We can therefore calculate the model's ideal code length and other information-theoretic metrics based on the protocol.

Theorem 3.4 (Ideal Code Length). *Consider a finite parameter model $p_{\boldsymbol{\theta}}(\cdot)$ training on dataset $\mathcal{D}$, the ideal code length $\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x})$ of a random response $\boldsymbol{x}$ compressed by $p_{\boldsymbol{\theta}}$ can be expressed as:*

$$\mathbb{E}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x})\right] = \left\lceil \frac{|\boldsymbol{x}|}{d} \right\rceil \left\lceil -\sum_{l=1}^{d} \sum_{j=1}^{2^{l-1}} p_{lj} \log p_{lj} \right\rceil$$

where d represents the depth of the $\mathcal{T}_{\mathcal{D}}$ after pruning under Definition 3.3 protocol, and p_{lj} represents the probability values of the leaf nodes for the j-th node at the l-th layer.

Since training and alignment involve multiple datasets with different and independent distributions, we consider the joint compression scenario for multiple datasets. For N pairwise disjoint datasets $\mathcal{D}_1, \cdots, \mathcal{D}_N$, the node weights $p_{lj}^{\mathcal{D}}$ of the token tree for the dataset $\mathcal{D} = \bigcup_{i=1}^{N} \mathcal{D}_i$ satisfy:

$$p_{lj}^{\mathcal{D}} = \frac{\sum_{i=1}^{N} p_l^{\mathcal{D}_i} |\mathcal{D}_i|}{\sum_{i=1}^{N} |\mathcal{D}_i|},$$

where $p_{lj}^{\mathcal{D}_i}$ stands for the probability value for nodes in $\mathcal{T}_{\mathcal{D}_i}$ while $|\mathcal{D}_i|$ represents the size of $\mathcal{D}_i$. Thus, the compression rate $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i}$ on specific datasets can also be defined accordingly:

$$\begin{aligned}\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i} &= \mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}_i}\left[\frac{\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}_i}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i}(\boldsymbol{x})\right]}{|\boldsymbol{x}|}\right] \\ &= \Theta\left(-\frac{\sum_{l=1}^{d}\sum_{j=1}^{2^{l-1}} p_{lj}^{\mathcal{D}_i} \log p_{lj}^{\mathcal{D}}}{d}\right).\end{aligned}$$

Here, the compression rate is defined as the compressed encoding length divided by the original length (Delétang et al., 2023) and ensures consistency between the training and compression objective, which means that minimizing the training loss is equivalent to minimizing the compression rate. Please see Appendix A for more details.

3.3 The Formal Definition of *Elasticity*

To formalize *inverse alignment* and *elasticity*, we provide precise definitions of these concepts.

Definition 3.5 (*Inverse Alignment*). Given a language model $p_{\boldsymbol{\theta}_0}$, aligned on dataset $\mathcal{D}_a$ to produce the aligned model $p_{\boldsymbol{\theta}_1}$. For any $\epsilon > 0$, if applying a dataset $\mathcal{D}_b$ (where $|\mathcal{D}_b| \ll |\mathcal{D}_a|$) to $p_{\boldsymbol{\theta}_1}$ yields $p_{\boldsymbol{\theta}_0'}$ such that $\rho(p_{\boldsymbol{\theta}_0'}, p_{\boldsymbol{\theta}_0}) \leq \epsilon$ for a given eval metric ρ, we define the transition from $p_{\boldsymbol{\theta}_1}$ back to $p_{\boldsymbol{\theta}_0'}$ as *inverse alignment*.

Definition 3.6 (The *Elasticity* of LLMs). Consider a language model $p_{\boldsymbol{\theta}_0}$ and transformation $p_{\boldsymbol{\theta}_0} \xmapsto{f(\mathcal{D}_a)} p_{\boldsymbol{\theta}_1}$, *elasticity* is said to exist in $(p_{\boldsymbol{\theta}_0}, \mathcal{D}_a)$ if there is an algorithmically simple *inverse operation* g and a dataset $\mathcal{D}_b$ such that $|\mathcal{D}_b| \ll |\mathcal{D}_a|$, with the property that:

$$p_{\boldsymbol{\theta}_1} \xmapsto{g(\mathcal{D}_b)} p_{\boldsymbol{\theta}_0'} \text{ and } \rho(p_{\boldsymbol{\theta}_0'}, p_{\boldsymbol{\theta}_0}) \leq \epsilon_0.$$

where ϵ_0 is a constant.

The eval metrics ρ can be viewed as a measure of behavioral and distributional proximity between models. In the context of the compression theorem, we use compression rate $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i}$ as the metric ρ to evaluate *elasticity* during the alignment process.

4 Why *Elasticity* Affects Alignment?

By now, we hope to have convinced the reader of the concept of compression modeling in language models' training and alignment. In the following, we apply this to analyze language models' training and alignment process, focusing on the underlying reasons that lead the model to alignment resistance.

In Figure 1, we have already presented the theorem of *elasticity*: when subject to fine-tuning perturbations, language models tend to retain the distribution associated with larger datasets while rejecting that of smaller ones. But why *elasticity* resists alignment? Is there a corresponding *elastic* invariant? How does *elasticity* make *inverse alignment* possible? In this section, we will formalize *elasticity* of language models and analyze the impact of *elasticity* on the model's behavior.

4.1 Formal Derivation of *Elasticity*

Our primary goal is to investigate the behavioral changes of a language model after pre-training and alignment, particularly in response to perturbations, typically caused by fine-tuning with a minimal dataset. We use compression rate as an evaluation metric to assess these changes. In analyzing the model's behavior, we focus on two datasets: $\mathcal{D}_p$, a larger dataset representing pre-training or the primary objective of training, and $\mathcal{D}_a$, a smaller dataset representing the alignment process or the secondary objectives of training. Since the pre-training stage encompasses a wide range of data, without loss of generality, we assume that the datasets in the alignment process follow the same distribution as some subset of the pre-training data.

Definition 4.1 (Normalized Compression Rate)**.** For N distinct datasets $\mathcal{D}_1, \cdots, \mathcal{D}_N$ and a parameter model $p_{\boldsymbol{\theta}}$ compressing $\mathcal{D} = \bigcup_{i=1}^{N} \mathcal{D}_i$, the normalized compression rate $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i/\mathcal{D}}$ for a particular dataset $\mathcal{D}_i$ is defined as:

$$\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i/\mathcal{D}} = \gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_i} - \log M, \tag{1}$$

where M is the number of leaf nodes of the pruned tree $\mathcal{T}_i'$ of dataset $\mathcal{D}_i$.

The normalized compression rate allows for comparing the model's compression performance across different datasets. The smaller the normalized compression rate of a dataset, the better the model's compression performance on the dataset.

With this definition in hand, we proceed to our main result: language models exhibit *elasticity*.

Theorem 4.2 (*Elasticity* of Language Models)**.** *Consider the pre-training dataset $\mathcal{D}_p = \bigcup_{i=1}^{3} \mathcal{D}_i$, the alignment dataset $\mathcal{D}_a$, and the perturbation dataset $\mathcal{D}_t$, with the model $p_{\boldsymbol{\theta}}(\cdot)$ trained on $\mathcal{D} = \mathcal{D}_p \cup \mathcal{D}_a \cup \mathcal{D}_t$. Assume that $\mathcal{D}_a \overset{d}{\sim} \mathcal{D}_2$, $\mathcal{D}_t \overset{d}{\sim} \mathcal{D}_3$, and $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$ are each distributed according to a Pareto mass distribution (Newman, 2005). As the perturbation data volume $|\mathcal{D}_t|$ varies, $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}$ and $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}$ satisfy:*

$$\begin{aligned} &\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} = \Theta\left(-k\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}\right), \\ &\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l} < 0, \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} > 0, \end{aligned} \tag{2}$$

where $l = \frac{|\mathcal{D}_t|}{|\mathcal{D}_a|} \ll 1$, $k = \frac{|\mathcal{D}_p|}{|\mathcal{D}_a|} \gg 1$, and $\{\mathcal{D}_i\}_{i=1}^{3}$ are datasets of equal cardinality.

Theorem 4.2 shows that as the perturbation increases, the normalized compression rates of the model for $\mathcal{D}_p$ decrease and $\mathcal{D}_a$ increase and the rate of changes is strongly correlated with the size of the datasets. Unlike the proportional changes in compression rates across different datasets, the language model seems to *prefer* the dataset with a larger volume, leading to biased model behavior after the perturbation.

4.2 *Elasticity* and *Inverse Alignment*

Theorem 4.2 shows the normalized compression rate change of different datasets is inversely proportional to their sizes under perturbations. A significant size difference between the pre-training and alignment datasets causes rapid performance degradation on the alignment dataset due to the inverse relationship, often spanning several orders of magnitude.

Intuitively, the model's compression process across multiple datasets is akin to resource allocation between towns: in a region with both a large metropolis and rural villages, to maximize overall economic productivity, resources are typically allocated to the metropolis first, to leverage its scale and agglomeration effects. In other words, the metropolis/large dataset occupies a dominant position in the system.

Therefore, the *elasticity* of language models makes *inverse alignment* possible. Due to the substantial data volume disparity between pre-training and alignment datasets, the model tends to revert to the pre-training rather than alignment distribution when subsequent perturbations occur. This indicates an inherent tendency toward *inverse alignment* during fine-tuning. Maximizing the impact of *elasticity* through well-designed perturbations will be key to achieving true *inverse alignment*.

4.3 *Elasticity* and the Hooke's Law.

The inverse proportionality result in Theorem 4.2 provides a potential invariant in the model training and alignment process: after perturbation, the rate of change in the compression rates of different datasets is inversely proportional to their sizes, with the absolute value of the product being a constant. This constant characterizes the impact of the perturbation on the model and indirectly describes the model's resistance to perturbations, or its *elasticity*.

The *elasticity* of the model can be intuitively analogized to a series system of springs, as shown in Figure 1. Consider two massless springs in series, with spring constants k_1 and k_2, respectively. When the entire system undergoes deformation due to an external force F, the system reaches a stable state, and the elastic force exerted by each spring is equal to F. According to Hooke's Law (Hooke, 2016), the elongation Δl_1 and Δl_2 of each spring is inversely proportional to its spring constant. Thus, in this system, we have:

$$F \propto k_1 \cdot \Delta l_1 = k_2 \cdot \Delta l_2 \ .$$

In the language model setting, after integrating Theorem 4.2 to l, we obtain $\Delta \gamma_{p_{\theta}}^{\mathcal{D}_i/\mathcal{D}}$ across different datasets, which is equivalent to the change in the KL divergence $\Delta D_{\mathrm{KL}}(\mathcal{P}_{p_{\theta}}||\mathcal{P}_{\mathcal{D}_i})$ between the model's distribution and the distributions of the individual datasets, is inversely proportional to the size of the datasets $|\mathcal{D}_i|$. Here, we only consider the absolute value of ΔD_{KL}. Analogous to the series spring model, the *elasticity* F in LLMs satisfies:

$$F \propto |\mathcal{D}_i| \cdot \Delta D_{\mathrm{KL}}(\mathcal{P}_{p_{\theta}}||\mathcal{P}_{\mathcal{D}_i}) \ , \tag{3}$$

where ΔD_{KL} corresponds to Δl in the spring model, while $|\mathcal{D}|$ corresponds to the spring constant k, thus leading to the *elasticity* of LLMs.

5 How *Elasticity* Resists Alignment?

In the previous sections, we proved that LLMs have *elasticity*. This section will analyze two specific phenomenons of it:

- ***Resistance* for Pre-Trained Models.** Models tend to maintain the original distribution and resist alignment;
- ***Rebound* for Post-Trained Models.** Fine-tuning in the opposite direction of post-training (*e.g.*, safe *vs.* unsafe) causes post-trained models to return quickly to the pre-training distribution.

5.1 Existence of Language Models' *Resistance*

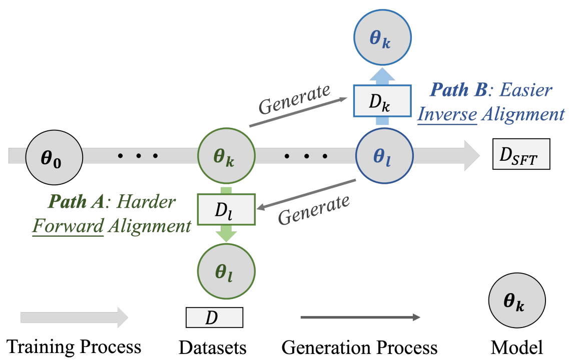

Figure 2: **Experiment pipeline for validating *resistance*.** We conceptualize *resistance* as: ***inverse alignment*** is easier than ***forward alignment***.

Experiment Design. We verify the existence of *resistance* by arguing that ***forward alignment*** is harder than ***inverse alignment*** for pre-trained models. Specifically, we first perform one epoch of SFT on a pre-trained LLM with parameters $\boldsymbol{\theta}_0$, saving the slices $\{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots, \boldsymbol{\theta}_n\}$. Subsequently, without loss of generality, we collect the responses of slices $\boldsymbol{\theta}_k$ and $\boldsymbol{\theta}_l$ (where $k < l$) on hold-out prompts, forming datasets D_k and D_l. As shown in Figure 2, we define ***forward alignment* (*Path A*)** as the process of training $\boldsymbol{\theta}_k$ on D_l, and ***inverse alignment* (*Path B*)** as the process of training $\boldsymbol{\theta}_l$ on D_k.

Experiment Setup. We select different SFT datasets including Alpaca (Taori et al., 2023), TruthfulQA (Lin et al., 2022), and Beavertails (Ji et al., 2024c,b), which correspond respectively to the model's 3H principle (Askell et al., 2021). We divide them into three equal parts to obtain three corresponding slices $\{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3\}$. We consider Llama2-7B, Llama2-13B, and Llama3-8B (Touvron et al., 2023) as the base model $\boldsymbol{\theta}_0$.

Experiment Results. As shown in Table 1, the experimental results indicate that the training loss of ***inverse alignment*** consistently remains lower than that of ***forward alignment***, irrespective of the slice pair selection. This observation holds across all models and datasets in the experiments. All experimental results demonstrate that ***inverse alignment*** is easier than ***forward alignment*** across diverse models and datasets and validate the existence of *resistance*. To further verify the *resistance* in language models, we provide additional experimental details and results in the Appendix B.1.

5.2 Existence of *Rebound*

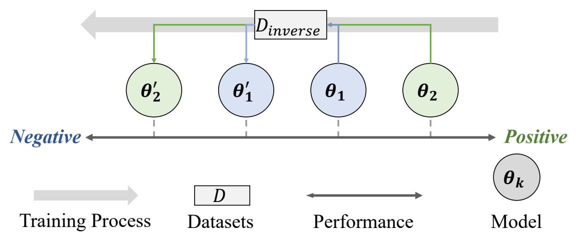

Figure 3: **Experiment pipeline for validating *rebound*.** We conceptualize *rebound* as: the more **positive** the post-trained models' performance, the more **negative** it becomes after inverse finetuning.

Experiment Design. We verify the existence of *rebound* by arguing that for post-trained models,